

S04 REVERBERATION TIME | O (MAX : 2 PT)

Intent : Design spaces in accordance with comfortable reverberation times that support speech intelligibility, vocal effort and are conducive to concentration.

Summary : This WELL feature requires that steps be taken to address acoustical comfort, by controlling reverberation time based on room functionality.

Issue : Reverberation time is the length of time taken for a sound to decay by 60 dB from an initial impulse level. The length of reverberation time is a function of room size, surface area and sound absorbing properties of surface finishes. Spaces with longer reverberation times may be larger in volume with hard surfaces that reflect sound. Spaces with shorter reverberation may be smaller with softer surfaces that absorb sound. Spaces with high reverberation may have increased ambient noise levels and reduce speech intelligibility (e.g., public address, speech reinforcement and amplified speech). Studies have shown that high reverberation times in classrooms increase auditory workload in students and reduce cognition, memory retention and concentration.¹⁻⁴ Similarly, studies have shown that high reverberation reduces speech intelligibility among hard of hearing and non-native speaking populations.⁵⁻⁸

Solutions : Reverberation time can be controlled by adding absorptive surface finishes at ceilings, walls and furniture.⁹⁻¹¹ Projects that can alter room geometry can change layouts and room dimensions to support optimal reverberation times, as needed. Reducing reflective surfaces, such as glass, drywall, stone or similar, will also reduce reflected sound energy, which increases reverberation time. Reducing reverberation time also allows audio equipment for telecommunication/AV, speech reinforcement or public address to operate with higher speech intelligibility performance.^{5,12-14}

PART 1 ACHIEVE REVERBERATION TIME THRESHOLDS (MAX : 2 PT)

For All Spaces except Dwelling Units & Guest Rooms:

Option 1: Reverberation time, design

For projects in which the room types listed in the table cumulatively make up at least 10% of occupiable project area, the following requirements are met:

- a. Reverberation time is within the ranges shown in the following table:

Room Type	Room Volume, v (cubic meters)	Reverberation Time, t (seconds) ^{5,9,10}
	$v < 280 \text{ m}^3$	$t \leq 0.6$
Areas for learning	$280 \text{ m}^3 \leq v \leq 570 \text{ m}^3$	$t \leq 0.8$
Areas for lectures		
Areas for conferencing	$v > 570 \text{ m}^3$	$t \leq 1.0$
Areas with regularly used PA systems	N/A	$t \leq 1.5$
Areas for dining	N/A	$t \leq 1.0$
	$v < 280 \text{ m}^3$	$t \leq 0.8$
Areas for fitness	$280 \text{ m}^3 \leq v \leq 570 \text{ m}^3$	$t \leq 1.1$
	$v > 570 \text{ m}^3$	$t \leq 1.8$
	$v < 280 \text{ m}^3$	$t \leq 1.1$
Areas for music rehearsal	$280 \text{ m}^3 \leq v \leq 570 \text{ m}^3$	$t \leq 1.4$

Note : Where room types include multiple use types (e.g., learning and fitness) use the limits that include the lower reverberation time or range.

OR

Option 2: Reverberation time, performance

For projects in which the room types listed in the table cumulatively make up at least 10% of occupiable project area, the following requirements are met:

- a. Reverberation time is within the ranges shown in the following table:

Room Type	Room Volume, v (cubic meters)	Reverberation Time, t (seconds) ^{5,9,10}
Areas for learning	$v < 280 \text{ m}^3$	$t \leq 0.6$
Areas for lectures	$280 \text{ m}^3 \leq v \leq 570 \text{ m}^3$	$t \leq 0.8$
Areas for conferencing	$v > 570 \text{ m}^3$	$t \leq 1.0$
Areas with regularly used PA systems	N/A	$t \leq 1.5$
Areas for dining	N/A	$t \leq 1.0$
	$v < 280 \text{ m}^3$	$t \leq 0.8$

Areas for fitness	$280 \text{ m}^3 \leq v \leq 570 \text{ m}^3$	$t \leq 1.1$
	$v > 570 \text{ m}^3$	$t \leq 1.8$
Areas for music rehearsal	$v < 280 \text{ m}^3$	$t \leq 1.1$
	$280 \text{ m}^3 \leq v \leq 570 \text{ m}^3$	$t \leq 1.4$

Note : Where room types include multiple use types (e.g., learning and fitness) use the limits that include the lower reverberation time or range. Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

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